CMAN :=

Given data, N=32, FP32 = 4 bytes, DRAM bandwidth = 320GB/s
compute = 10TFLOPS.

Task 1:=

N=32 times, so each B[K][J] accessed 32 times

A access $N^3 = 32^3 = 32,768$

B access $N^3 = 32^3 = 32,768$

Total elements access $= 2N^3 = 2 \times (32)^3 = 65,536$

Total DRAM units $= 2N^3 \times 4 = 65,536 \times 4$

$= 262,144$ bytes $\simeq 256KB$

$\boxed{\text{Total DRAM traffic} = 256KB}$

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Task 2 :-

Tiled size $T=8$

$\left(\frac{N}{T}\right)^2 = \left(\frac{32}{8}\right)^2 = 16$ tiles per matrix

Each tile holds $(T^2) = (8)^2 = 64$ elements

$= 64 \times 4$ bytes

$= 256$ bytes

Dram loads for $A = N^2 \times 4$

$= (32)^2 \times 4 = 4096$ bytes

total Dram for A and B

$= N^2 \times 4 \times 2$

$= 4096 \times 2 = 8192$ bytes

Dram traffic = total Dram load for A and B is 8192 bytes.

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Task 3 :=

Traffic ratio :=

Naive traffic = 262,144 bytes
Tiled traffic = 32,768 bytes

$$\text{Traffic ratio} = \frac{\text{Naive}}{\text{Tiled}} = \frac{262,144}{32,768} = 8 \ (T)$$

Tiled traffic = Naive traffic ÷ N

= 262,144 ÷ 32

Tiled traffic = 8,192 bytes

$$\text{Ratio} = \frac{262,144}{8,192} = 32 = N$$

- In the naive traffic, we reload matrix B from memory 32 times because every row of A needs to read all of B again. Tiling fixes this by keeping B in fast memory and reusing it, so we only load B once. That is why native method uses 32 times more memory traffic exactly N times more.

Task 4 :=

Bandwidth = 320GB/s
compute = 10TFLOPS
Total FLOPs = $2 \times N^3 = 2 \times 32^3 = 65,536$ FLOPs

Naive case :=

t_memory = Traffic / BW

= 2,62 144 / (320 GB/s)

$= 8.192 \times 10^{-7} s = 0.819 \mu s$

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$t_compute = FLOPs / P$

$= 65{,}536 / (1 \times 10^{13})$

$= 6.554 \times 10^{-9} s = 0.00655\ \mu s$

$t_memory\ (0.819\ \mu s) >> t_compute\ (0.00655\ \mu s)$

$\boxed{t_exec\ (naive) = 0.819\ \mu s}$

Bottleneck : Memory - bound.

Tiled case (T=8) :=

$t_Memory = Traffic / BW$

$= 32{,}768 / (3.2 \times 10^{11})$

$= 1024 \times 10^{-7} s = 0.1024\ \mu s$

$t_compute = FLOPs / P$

$= 65{,}536 / (1.0 \times 10^{13})$

$= 6.554 \times 10^{-9} s = 0.00655\ \mu s$

$t_memory\ (0.1024\ \mu s) >> t_compute\ (0.00655\ \mu s)$

$\boxed{t_exec\ (tiled) = 0.1024\ \mu s}$

Bottleneck : Memory- bound

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